Type MS · Scale per-unit or reference quantities and combine them

Answer 1 of 10 · Question 2 part a · 3 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

2 (a)	• $\frac{6.1}{100}$; • x85; • 5.2;	Allow ecf for mps 2 and 3 If 5.18 given for mp3 then max 2 marks Full marks for correct final answer	3
(b)	• add sample to boiling tube; • add biuret solution; • purple if present, (remains blue if absent);		3
(c)	• $\frac{100}{16}$; • = 6.25/7;	ecf 1 mark max for correct final answer to a division calculation	2
(d)	Any five from: • contains calcium; • contains vitamin D; • vitamin D allows calcium uptake/absorption (from intestine); • calcium absorbed/stored in bones/used to form compact bone; • to strengthen bones/increases bone density; • also contains protein which is required for (bone) development/growth;	Allow makes bones less brittle	5

Total 13

Type MS · Scale per-unit or reference quantities and combine them

Answer 1 of 10 · Question 2 part a · 3 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 2 of 10 · Question 8 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

8(a)	<table><tr><th rowspan="2">Organ</th><th rowspan="2">Mass in kg</th><th colspan="2">Rate of blood flow in cm³ per minute</th></tr><tr><th>through the whole organ</th><th>per 100g of the organ</th></tr><tr><td>kidneys</td><td>0.3</td><td>1200</td><td>400</td></tr><tr><td>skeletal muscles</td><td>25.0</td><td>750</td><td>3</td></tr><tr><td>liver</td><td>1.5</td><td>1500</td><td>100</td></tr></table> kidney 1200÷3; 400; skeletal muscle 250 × 3; 750;	correct answer award full marks Ecf for each calculation allow correct answer to wrong figures	**(4)**

Question Number	Answer	Additional guidance	Mark
8(b)	Any four from: • (receives blood from) hepatic artery; • (receives blood from) hepatic portal vein; • homeostasis; • reference to blood sugar regulation; • detoxification (of alcohol)/urea formation/deamination;	For mp 1 and 2 allow receives blood from two vessels for one mark	**(4)**

Type MS · Scale per-unit or reference quantities and combine them

Answer 2 of 10 · Question 8 part a · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
8(c)	Any four from: • less blood flow to kidney/liver; • increased blood flow to skeletal muscles; • (muscles have) greater energy demand/more energy needed/to reduce oxygen debt/lactic acid; • (so) more oxygen/glucose needed; • for (aerobic) respiration;		**(4)**

Total 12 Marks

Paper Total 90 Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 3 of 10 · Question 3 part a.v · 4 marks

2024 JUN · Paper 01 (regional R) · find part a.v in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 4 of 10 · Question 5 part e · 3 marks

2024 JUN · Paper 02 (regional R) · find part e in this mark scheme extract

5	(a)	(cardiac) muscle(s);		(1)
	(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
	(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
	(d) (i)	hormones;		(1)
	(ii)	platelets/fibrinogen/prothrombin;		(1)
	(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 5 of 10 · Question 7 part a.iii · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

7 (a) (i)	Any two from the following • causes cancer; • gene mutation; • incorrect protein made/shape of protein incorrect;		2
(ii)	• changing the way free radicals behave; • to allow normal flow of substances/named substance in and out of cells;		2
(iii)	• 40g = 60% of daily requirement; • $\frac{40 \times 100}{60}$; • = 67g;	Allow 66.7g	3
(b)	• add fresh orange juice to a test tube/named suitable container; • add DCPIP drop by drop until it no longer decolourises; • record the volume of DCPIP added; • repeat with the same volume of boiled orange juice; • compare volumes;		5
			Total 12

Type MS · Scale per-unit or reference quantities and combine them

Answer 5 of 10 · Question 7 part a.iii · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 6 of 10 · Question 2 part b.ii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a)		• all components/all nutrients/all required nutrients/all food groups/protein, carbohydrates, fats, vitamins, minerals ; • in the correct quantity/correct proportion/ recommended intake;		2
(b) (i)		Protein/ /vitamins/named vitamin/minerals / named mineral/starch/fibre;		1
	(ii)	• $\frac{100}{4} = 25$; • 25×2.4; • = 60 g; **60 = 3 marks**	ECF for mp 2 and 3	3
	(iii)	• 4 slices = 470 (kJ)/1 slice = 118 (kJ); • $\frac{11\,800}{118} = 100.4$ slices; ***2 marks for 100.4 or 101or 100.43***	1 mark for 11800÷470 **or** 1 mark for 100	2
	(iv)	• break up/chop meat/ liquidise/small pieces; • place in test/boiling tube; • add alcohol/ethanol; • add water; • milky/cloudy/ milky white colouration shows presence of fat;	Apply the marking points independently Award mp3 and mp4 in any order. Accept mp3 and 4 in any order	5

Total = 13 marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 6 of 10 · Question 2 part b.ii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 7 of 10 · Question 2 part c · 2 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

2 (a)	• $\frac{6.1}{100}$; • x85; • 5.2;	Allow ecf for mps 2 and 3 If 5.18 given for mp3 then max 2 marks Full marks for correct final answer	3
(b)	• add sample to boiling tube; • add biuret solution; • purple if present, (remains blue if absent);		3
(c)	• $\frac{100}{16}$; • = 6.25/7;	ecf 1 mark max for correct final answer to a division calculation	2
(d)	Any five from: • contains calcium; • contains vitamin D; • vitamin D allows calcium uptake/absorption (from intestine); • calcium absorbed/stored in bones/used to form compact bone; • to strengthen bones/increases bone density; • also contains protein which is required for (bone) development/growth;	Allow makes bones less brittle	5

Total 13

Type MS · Scale per-unit or reference quantities and combine them

Answer 7 of 10 · Question 2 part c · 2 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 8 of 10 · Question 2 part b.iii · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

2 (a)		• all components/all nutrients/all required nutrients/all food groups/protein, carbohydrates, fats, vitamins, minerals ; • in the correct quantity/correct proportion/ recommended intake;		2
(b) (i)		Protein/ /vitamins/named vitamin/minerals / named mineral/starch/fibre;		1
	(ii)	• $\frac{100}{4} = 25$; • 25×2.4; • = 60 g; **60 = 3 marks**	ECF for mp 2 and 3	3
	(iii)	• 4 slices = 470 (kJ)/1 slice = 118 (kJ); • $\frac{11\,800}{118} = 100.4$ slices; ***2 marks for 100.4 or 101or 100.43***	1 mark for 11800÷470 **or** 1 mark for 100	2
	(iv)	• break up/chop meat/ liquidise/small pieces; • place in test/boiling tube; • add alcohol/ethanol; • add water; • milky/cloudy/ milky white colouration shows presence of fat;	Apply the marking points independently Award mp3 and mp4 in any order. Accept mp3 and 4 in any order	5

Total = 13 marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 8 of 10 · Question 2 part b.iii · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 9 of 10 · Question 7 part a.iv · 3 marks

2022 JAN · Paper 01 · find part a.iv in this mark scheme extract

Number					
7	(a)	(i)	carbon hydrogen and oxygen;		1
		(ii)	nitrogen	Allow sulfur	1
		(iii)	19.6 - 12; 7.6 g	Allow ECF	2
		(iv)	protein 1.2 g × 17 = 20.4 kJ; lipid 4.8 g × 37 = 177.6 kJ; total = 198 kJ;	Allow ECF for 2nd and 3rd mark	3
	(b)		add alcohol to fish; add water; cloudy/milky appearance/emulsion indicates lipid;		3

Total 10 marks

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Type MS · Scale per-unit or reference quantities and combine them

Answer 10 of 10 · Question 6 part b.i · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Scale per-unit or reference quantities and combine them

Answer 10 of 10 · Question 6 part b.i · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	